Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Computer Application

MCA | Sem-3 | Winter-2025

Course Code	:	2305CS332	**Date**	:	08-10-2025
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Explain various delay which occur in data packet transmission.	**4**
	(B)	What is topology? Explain any one topology in brief.	**3**
		OR	
		Give difference between LAN, MAN and WAN.	
	(C)	Discuss the circuit switching versus packet switching approaches for moving data through a network of links and switches.	**7**
		OR	
		Draw and explain different layers of OSI Model.	
Q.2	**(A)**	Explain the working of DHCP protocol.	**4**
	(B)	Differentiate HTTP GET method and POST method.	**3**
		OR	
		Compare persistent HTTP and non-persistent HTTP.	
	(C)	Explain SMTP protocol in detail.	**7**
		OR	
		Explain the hierarchical DNS system.	
Q.3	**(A)**	Compare UDP and TCP.	**4**
	(B)	Explain slow start component of TCP congestion control mechanism.	**3**
		OR	
		Provide the Port number associated with the following Application Layer Services: 1. SMTP 2. HTTP 3. DNS	
	(C)	Explain the various fields of TCP header.	**7**
		OR	
		Describe Go Back N and Selective Repeat protocol.	

Q.4	**(A)**	What is the relevance of Type of Service (ToS) and Time to Live (TTL) field in IPV4 packet?	**4**
	(B)	Compare IPv4 and IPv6.	**3**
		OR	
		Compare Datagram and Virtual Circuit.	
	(C)	Explain distance vector routing with suitable example.	**7**
		OR	
		Explain Link state routing with suitable example.	
Q.5	**(A)**	Explain CSMA/CD Protocol.	**4**
	(B)	Compare multicasting and multiple unicasting.	**3**
		OR	
		Draw and explain Ethernet Frame Structure.	
	(C)	What do you mean by random access protocols? Explain slotted ALOHA in brief.	**7**
		OR	
		Generate CRC code for message 1101010101. Given generator polynomial is x^4+x^2+1.	

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